

Customer Shopping Behaviour
CCDS 321 – Data Mining
Milestone 2

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Dataset Description

This dataset describes customer shopping behavior, with 5,050 observations and 17 attributes, which contains 13 categorical and 4 numerical, covering customer information in detail such as age, gender, and location. In addition, it contains purchase details such as category, item purchased, and purchase amount, as well as behavioral attributes such as subscription status, purchase frequency, and payment method.

Attribute Types

The dataset contains a mix of numerical and categorical variables. Numerical variables (Type: Num) represent measurable values and can be used for statistical analysis. These include variables such as **Age**, **Customer ID**, **Previous Purchases**, **Purchase Amount (USD)**, and **Review Rating**. On the other hand, categorical variables (Type: Char) represent descriptive information and are divided into different categories. These include variables such as **Category**, **Color**, **Gender**, **Item Purchased**, **Location**, **Payment Method**, **Season**, **Shipping Type**, **Size**, and **Subscription Status**. Categorical variables are mainly used to group data and analyze patterns between different customer segments, while numerical variables help in measuring and comparing values. This combination of variable types makes the dataset suitable for both classification and clustering analysis.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
2	Age	Num	8	BEST12.	BEST32.
5	Category	Char	11	\$11.	\$11.
9	Color	Char	7	\$7.	\$7.
1	Customer ID	Num	8	BEST12.	BEST32.
14	Discount Applied	Char	3	\$3.	\$3.
17	Frequency of Purchases	Char	14	\$14.	\$14.
3	Gender	Char	6	\$6.	\$6.
4	Item Purchased	Char	10	\$10.	\$10.
7	Location	Char	14	\$14.	\$14.
16	Payment Method	Char	13	\$13.	\$13.
15	Previous Purchases	Num	8	BEST12.	BEST32.
6	Purchase Amount (USD)	Num	8	BEST12.	BEST32.
11	Review Rating	Num	8	BEST12.	BEST32.
10	Season	Char	6	\$6.	\$6.
13	Shipping Type	Char	14	\$14.	\$14.
8	Size	Char	2	\$2.	\$2.
12	Subscription Status	Char	3	\$3.	\$3.

Data Cleaning

1. Missing Data Handling

Overview

The dataset contains 5,050 customer records. An initial assessment was conducted to identify missing values across all variables. After identifying the missing data, appropriate imputation methods were applied to handle missing values while preserving the original data distribution as much as possible.

Missing Data Frequencies		
Legend: ., A, B, etc = Missing		
Customer ID	Frequency	Percent
Non-missing	5050	100.00
Age	Frequency	Percent
Non-missing	5050	100.00
Gender	Frequency	Percent
Non-missing	5050	100.00
Item Purchased	Frequency	Percent
Non-missing	5050	100.00
Category	Frequency	Percent
Non-missing	5050	100.00
Purchase Amount (USD)	Frequency	Percent
.	556	11.01
Non-missing	4494	88.99
Location	Frequency	Percent
Non-missing	5050	100.00
Size	Frequency	Percent
	370	7.33
Non-missing	4680	92.67
Color	Frequency	Percent
Non-missing	5050	100.00
Season	Frequency	Percent
Non-missing	5050	100.00
Review Rating	Frequency	Percent
.	601	11.90

Results: Describe Missing Data		
Review Rating	Frequency	Percent
Non-missing	4449	88.10
Subscription Status	Frequency	Percent
Non-missing	5050	100.00
Shipping Type	Frequency	Percent
Non-missing	5050	100.00
Discount Applied	Frequency	Percent
Non-missing	5050	100.00
Previous Purchases	Frequency	Percent
.	548	10.85
Non-missing	4502	89.15
Payment Method	Frequency	Percent
Non-missing	5050	100.00
Frequency of Purchases	Frequency	Percent
Non-missing	5050	100.00

The analysis shows that most variables in the dataset are complete and do not contain missing values. However, a few variables have some missing data that needs to be handled.

Missing values were found in the following variables:

Purchase Amount (USD): 11.01% missing

Review Rating: 11.90% missing

Previous Purchases: 10.85% missing

Size: 7.33% missing

All other variables have no missing values (100% complete).

Imputation Methods Used

Two different methods were applied based on variable type:

1-Numerical Variables (Purchase Amount, Review Rating, Previous Purchases)

Method Used: Mean Imputation

Why: Mean imputation is a standard method for handling missing numerical data. It replaces missing values with the mean of the available values for that variable.

Code Used:

```
proc stdize data=WORK.IMPORT
    out=WORK.FINAL_FIXED
    reponly
    method=mean ;
var 'Purchase Amount (USD)'n
    'Review Rating'n
    'Previous Purchases'n;
run;
```

Note: The REONLY option was used to ensure that **only missing values were replaced**. All original non-missing values remained unchanged.

2- Categorical Variable (Size)

Method Used: Mode Imputation (Most Frequent Value)

Why: For categorical variables, the mode (most frequent value) is the appropriate measure of central tendency.

Finding the Most Frequent Size:

A frequency analysis was performed on the Size variable:

Results: Program 1				
Size	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	370	7.33	370	7.33
M	1981	39.23	2351	46.55
L	1282	25.39	3633	71.94
S	874	17.31	4507	89.25
XL	543	10.75	5050	100.00

Result: The most frequent size was "**M**" with 1,981 occurrences (39.23%).

Code Used:

```
data WORK.FINAL_FIXED;  
  set WORK.FINAL_FIXED;  
  if missing(Size) then Size = "M";  
run;
```

Final Dataset Summary

After applying the imputation methods:

- All missing values were successfully replaced.
- Total rows: 5,050
- Total columns: 17
- No rows were deleted.
- All non-missing original values remained unchanged.

2. Duplicate Records Handling

Overview

The dataset was examined for fully duplicate records. A duplicate row is defined as a row where **all 17 attributes** are identical to another row in the dataset.

Detection Method

Tool Used: The Sort step in SAS Studio with the Remove Duplicate Rows option enabled.

Why: The Sort step removes rows that are completely identical across all columns, ensuring that each customer record appears only once in the dataset.

Findings

The analysis revealed that the dataset contained **50 fully duplicate rows**, where every attribute was completely identical across each pair of records.

Result

After removing the duplicate records:

- All duplicate rows were successfully removed.
- Total rows before: 5,050
- Duplicate rows removed: 50
- Total rows after: 5,000
- All original non-duplicate records remained unchanged.



3. Inconsistent Values

Overview

A review of all variables was conducted to identify inconsistent values. Categorical variables were examined using **One-Way Frequencies in SAS Studio**, and numerical variables were examined using **Summary Statistics**.

Why: Summary Statistics provides a comprehensive overview of numerical variables, including minimum and maximum values, which helps identify any invalid or out-of-range values. One-Way Frequencies examines all unique values in each categorical column to detect any inconsistent labels.

Variable	Mean	Std Dev	Minimum	Maximum	N
Previous Purchases	25.2321204	13.6939557	0	50.0000000	5000
Review Rating	3.6675498	0.8132186	1.0000000	5.0000000	5000
Purchase Amount (USD)	145.5820882	261.1314071	10.1200000	1499.76	5000
Age	44.1880000	15.2721480	18.0000000	70.0000000	5000

The summary statistics confirmed that there were no invalid numerical values, such as negative ages or ratings outside the expected range:

- **Age:** min = 18, max = 70
- **Review Rating:** min = 1.0, max = 5.0
- **Previous Purchases:** min = 0, max = 50

However, **Purchase Amount (USD)** showed unusually high values, which were treated as possible outliers rather than inconsistency issues and will be addressed in the Outliers section.

For categorical variables, all columns were reviewed and found to be consistent, with the exception of two columns: **Season** and **Frequency of Purchases**, where different labels were used to represent the same meaning.

Season Column

- **Issue:** Two values representing the same season were found: **Fall** (975) and **Autumn** (299)

The FREQ Procedure

Season	Frequency
Autumn	299
Fall	975
Spring	1266
Summer	1221
Winter	1239

- **Method:** Recode Values in SAS Studio — **Autumn replaced with Fall**
- **Why:** Recode Values replaces different labels that represent the same meaning with a single unified label, ensuring consistency across the dataset for accurate analysis.
- **Result:** After recoding, the Season column now contains **4 unique** values **only: Fall, Spring, Summer, and Winter**, with no duplicate representations.

The FREQ Procedure

Season	Frequency
Fall	1274
Spring	1266
Summer	1221
Winter	1239

Frequency of Purchases Column

- **Issue:** Four values representing only two meanings were found: **Bi-Weekly** (547) and **Fortnightly** (542) both mean every two weeks, and **Quarterly** (927) and **Every 3 Months** (584) both mean every three months.

Frequency of Purchases	Frequency
Annually	572
Bi-Weekly	547
Every 3 Months	584
Fortnightly	542
Monthly	936
Quarterly	927
Weekly	892

- **Method:** Recode Values in SAS Studio — **Fortnightly** replaced with **Bi-Weekly**, **Every 3 Months** replaced with **Quarterly**.
- **Why:** Recode Values replaces different labels that represent the same meaning with a single unified label, ensuring consistency across the dataset for accurate analysis.
- **Result:** After recoding, the Frequency of Purchases column now contains **5 unique values only: Annually, Bi-Weekly, Monthly, Quarterly, and Weekly**.

Frequency of Purchases	Frequency
Annually	572
Bi-Weekly	1089
Monthly	936
Quarterly	1511
Weekly	892

Final Dataset Summary

After fixing the inconsistent values:

- All inconsistent values were successfully corrected.
- Total rows: 5,000
- Total columns: 17
- Season column: reduced from 5 to 4 unique values
- Frequency of Purchases column: reduced from 7 to 5 unique values

4. Outlier Detection and Removal

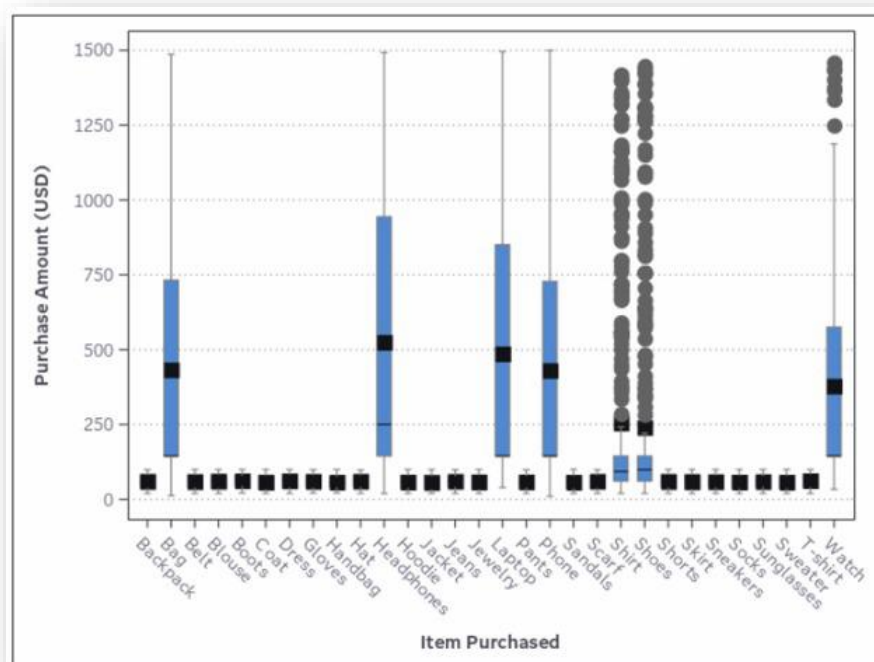
Overview

As noted in the Inconsistent Values section, the **Purchase Amount (USD)** variable showed unusually high values that required further investigation.

Detection Method

Tool Used: Box Plot in SAS Studio, grouped by Item Purchased

Why: Since different products naturally have different price ranges, comparing a laptop price to a shirt price would produce misleading results. Therefore, outliers were detected separately for each item type. The Box Plot automatically calculates the **Maximum Whisker** for each item — any value above it is flagged as an outlier.



Findings

The Box Plot revealed outliers in the following products:

- **Shirt:** purchase amounts reached up to \$1,419, which is abnormally high compared to the typical price range of this item.
- **Shoes:** purchase amounts reached up to \$1,448, which is abnormally high compared to the typical price range of this item.
- **Watch:** purchase amounts reached up to \$1,459, which is abnormally high compared to the typical price range of this item.

All other products showed no outliers, as their price ranges were consistent and within expected bounds.

Removal Method

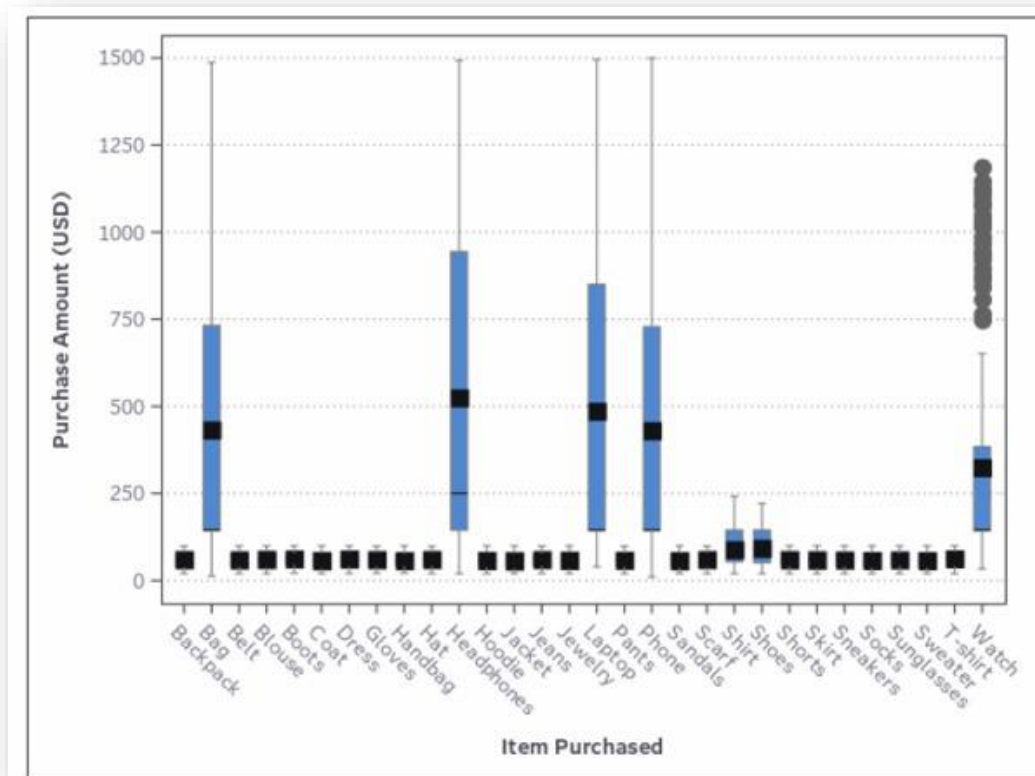
Tool Used: Filter Rows in SAS Studio

Why: Filter Rows allows keeping only the records that meet specific conditions, effectively removing any rows where the purchase amount exceeded the upper bound for each flagged item.

Three separate Filter Rows steps were applied:

- **Shirt:** removed all records where Purchase Amount > \$249
- **Shoes:** removed all records where Purchase Amount > \$249
- **Watch:** removed all records where Purchase Amount > \$1,187

Result



Note: After removing the outliers, the Box Plot recalculated the bounds based on the updated data. As a result, a few points in the **Watch** item may appear above the new whisker. However, these points were within the normal range before removal and do not represent significant outliers.

Final Dataset Summary:

- All identified outliers were successfully removed.
- Total rows before: 5,000
- Total outliers removed: 132
- Total rows after: 4,868
- Total columns: 17
- All non-outlier values remained unchanged.

Statistics

1. Numerical Attributes

The table below presents Descriptive measures for each attribute in the customer dataset. However, numeric attributes are measured by mean, median, and standard deviation. Meanwhile, variance hasn't been displayed in SAS, but it was calculated manually.

Filter		Descriptive Measures						Metadata Measures	Data Quality Measures
#	Name	Distinct Values	Mean	Median	Minimum	Maximum	Standard Deviation		
3	Customer ID	4,868	2466.609	2434.5	1	5099	1452.176		
4	Age	53	44.14544	44	18	70	15.27034		
8	Purchase Amount (USD)	494	124.9901	70	10.12	1499.76	224.784		
12	Review Rating	29	3.677002	3.668195	1	5	0.80238		
16	Previous Purchases	52	25.25218	25.2219	0	50	13.76855		
1	Frequency of Purchases	5	--	--	--	--	--		
2	Season	4	--	--	--	--	--		
5	Gender	3	--	--	--	--	--		
6	Item Purchased	30	--	--	--	--	--		
7	Category	5	--	--	--	--	--		

Variance calculations:

Age: 233.18

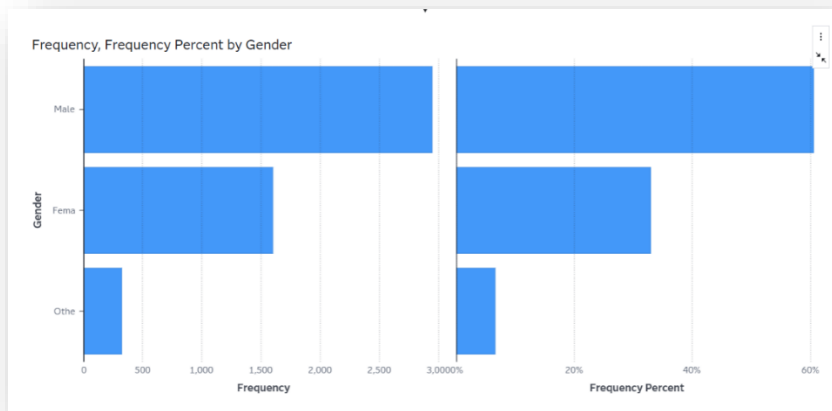
Purchase Amount: 50527.86

Review Rating:0.6438

Previous Purchases:189.57

2. Categorical Attributes

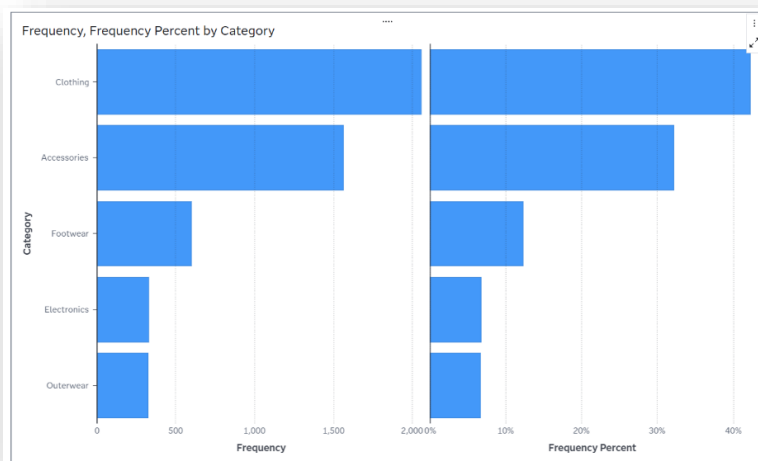
◆ Attribute: Gender



Interpretation

Most customers are Male (60.48%), followed by Female (32.89%). Which means Male is the most common gender in the dataset. Also, the mode of “Gender” is **Male**, because it has the highest frequency.

◆ Attribute: Category



Interpretation

Most purchases are Clothing (42.17%), followed by Accessories (32.11%). Which means customers buy Clothing the most in the dataset. Also, the mode of “Category” is **Clothing**, because it has the highest frequency.

◆ Attribute: Subscription Status



Interpretation

Most customers are No (68.73%), followed by Yes (31.27%). Which means No is the most common subscription status in the dataset. Also, the mode of “Subscription Status” is **No**, because it has the highest frequency.

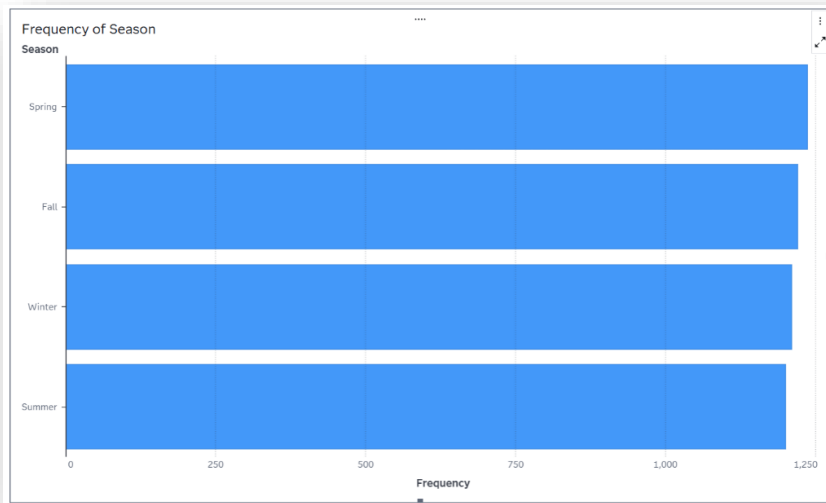
◆ Attribute: Payment Method



Interpretation

Most payments are Credit Card (18.69%), followed by Cash (18.65%). Which means the credit card is the most common payment method in the dataset. Also, the mode of “Payment Method” is **Credit Card**, because it has the highest frequency.

◆ Attribute: Season



Interpretation

Most purchases are in Spring (1,237), followed by Fall (1,221). Which means Spring is the most common season in the dataset. Also, the mode of “Season” is **Spring**, because it has the highest frequency.

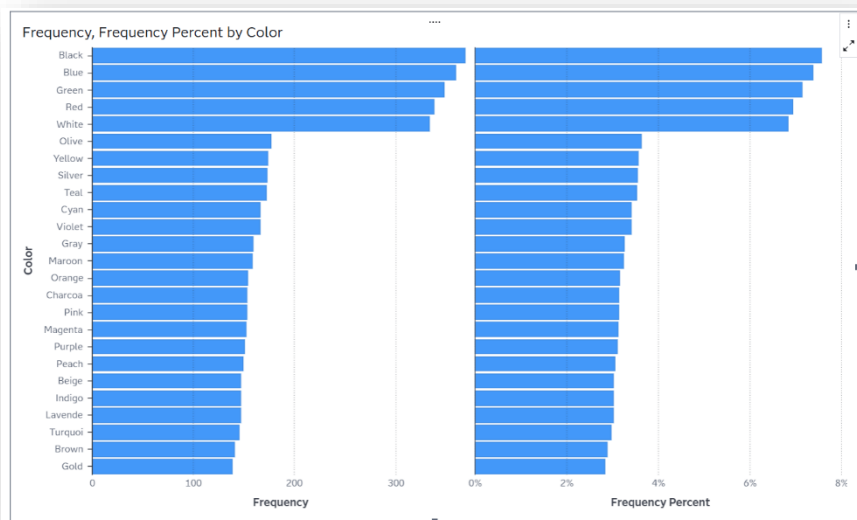
◆ Attribute: Shipping Type



Interpretation

Most orders use Standard (23.38%), followed by Express (23.21%). Which means Standard is the most common shipping type in the dataset. Also, the mode of “Shipping Type” is **Standard**, because it has the highest frequency.

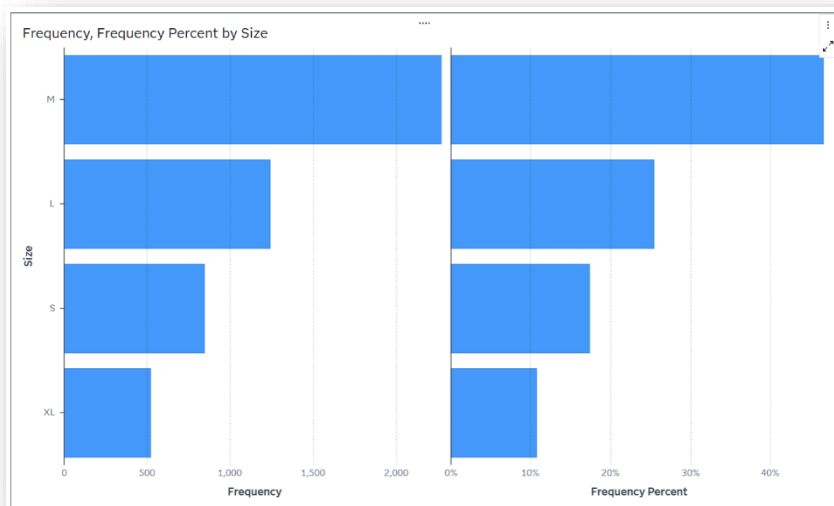
◆ Attribute: Color



Interpretation

Most orders use Black (7.56%), followed by Blue (7.37%). Which means Black is the most common color in the dataset. Also, the mode of “Color” is **Black**, because it has the highest frequency.

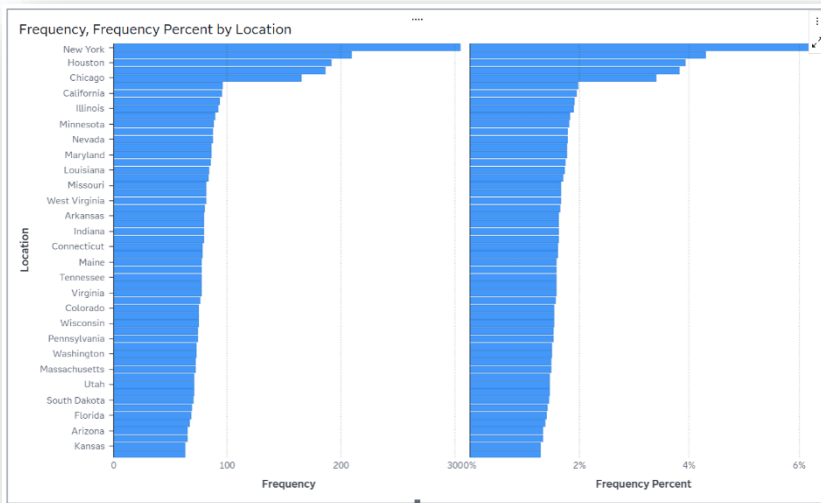
◆ Attribute: Size



Interpretation

Most orders use M (46.53%), followed by L (25.43%). Which means M is the most common size in the dataset. Also, the mode of “Size” is M, because it has the highest frequency.

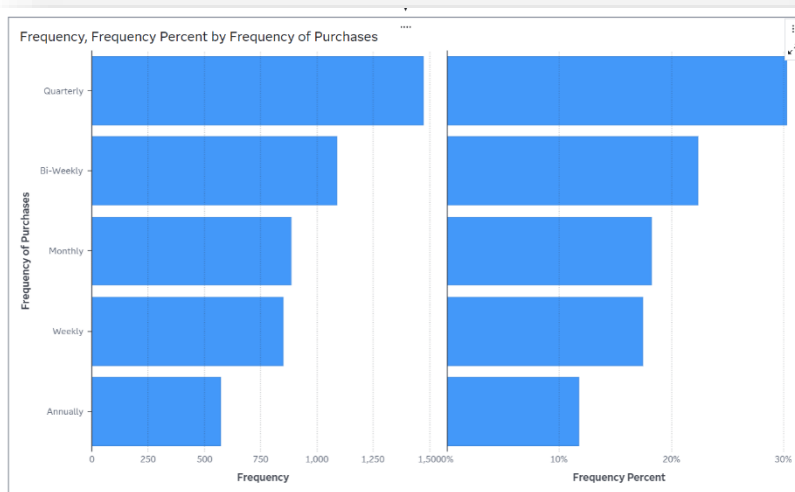
◆ Attribute: Location



Interpretation

Most orders use New York (6.24%), followed by Los Angeles (4.29%). Which means New York is the most common location in the dataset. Also, the mode of “Location” is New York, because it has the highest frequency.

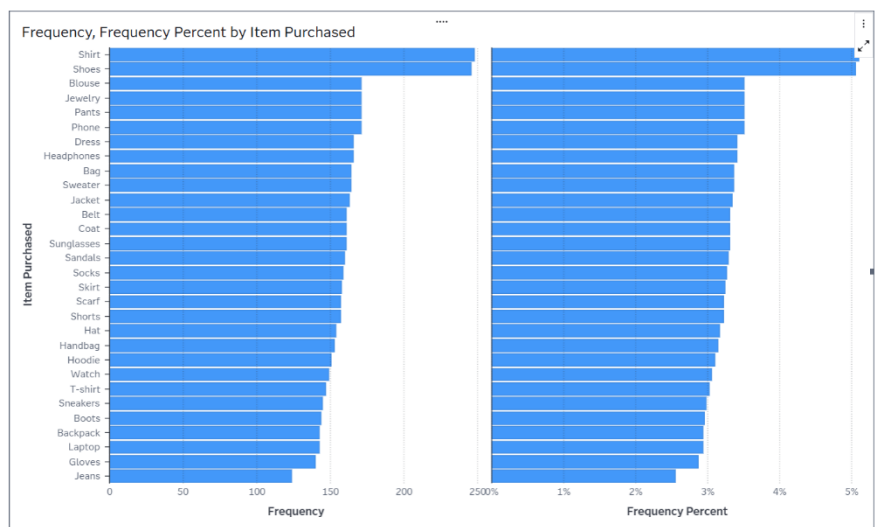
◆ Attribute: Frequency of Purchases



Interpretation

Most orders use Quarterly (30.22%), followed by Bi-Weekly (22.37%). Which means Quarterly is the most common frequency of purchases in the dataset. Also, the mode of “Frequency of Purchases” is Quarterly, because it has the highest frequency.

◆ **Attribute:** Item Purchased



Interpretation

Most orders use Shirt (5.09%), followed by Shoes (5.05%). Which means Shirt is the most common item purchased in the dataset. Also, the mode of Item Purchased is Shirt, because it has the highest frequency.